greenhouse effect. The role of CNES has been greatly enhanced by the establishment of the Space Coordination Group which is chaired by the Chief of Staff of the Armed Forces. CNES is an active and permanent member of the Group.

The **U.S.** and **Japan** reached an agreement providing for Earth observation data exchanges and distribution in global change studies, disaster prevention, environment monitoring and other related fields.

A recent agreement concluded by Aerospatiale, RKA and the Russian company TsSKb aims at using **Ariane** and **Soyuz** launch vehicles to put satellites from the Globalstar mobile vehicle communications system into orbit.

Tongasat, which licenses the use of orbital slots claimed by the Kingdom of Tonga, and Russia's **Intersputnik** have agreed to jointly operate a new communications satellite system to fill unused Tonga orbital slots.

The **ITU** World Radiocommunication Conference, held in Geneva, October-November 1995, reviewed possible far-reaching changes in the **radio regulations** which constitute an international treaty governing the use of the radio-frequency spectrum and satellite orbits. Among the approximately 200 new or revised recommendations, the Conference Assembly considered, for instance, the integration of terrestrial and satellite mobile communication systems and the protection of frequencies for radio astronomical measurements on the shielded side of the Moon. The Conference also debated the development of mobile satellite services, particularly those involving the little and big LEO's, as well as the allocation of radio-frequency spectrum for emerging new services, such as global personal communication systems.

ITU and **UNESCO** agreed to establish a joint pilot project to explore the use of distance learning in developing countries by applying interactive television.

The theme of **TELECOM 95** held in Geneva, October 3-11, 1995, was "CONNECT!," reflecting the trend of convergence currently taking place among major industries: telecommunications, information technology, and broadcasting. The TELECOM 95 forum consisted of two summits: the Strategy Summit, dealing with regulatory, policy, and financial issues; and the Technology Summit, focusing on developments, standardization and implementation of new technologies.

Minneapolis (USA) will be the site of the next ITU plenipotentiary conference and South Africa will host **TELECOM 98.**

Manfred Lachs Space Law Moot Court Competition

The final of the 4th Manfred Lachs Space Law Moot Court Competition was held in Oslo, Norway, October 6, 1995 between the teams of the University of North Carolina and Leiden University. It was won by North Carolina. The adjudicating judges were *G. Guillaume, V. Vereshchetin and Chr. Weeramantry* of the International Court of Justice. The case and written briefs will be published in the IISL Proceedings. Next year's final

will be held in Beijing during the IISL Colloquium (Oct. 7-12, 1996), and will deal with liability issues arising from commercial space activities.

Other Events

"Vision and Reality: Face to Face" was the theme of the 11th National Space Symposium in April 1995, which addressed many practical applications of space exploration and development, fiscal and political issues, as well as the challenges and potential of the future in space. Among exhibits of interest were several examples of spin-off technology, including a cool suit, a cordless drill, and excimer laser angioplasty.

The second symposium on the Moon and Other Celestial Bodies was held in Tokyo on September 1, 1995 under the sponsorship of NASDA and the Institute of Space and Astronautical Science (ISAS). Apart from the scientific aspects, there was a discussion of why and how we want to go to the moon in terms of cultural, economic and other aspects, and how Japan could contribute by promoting its lunar projects.

The Centre de Télédétection Spatiale de Toulouse (Center for Remote Sensing of Earth from Space of Toulouse) opened on September 25, 1995.

The 4th ECSL Practitioners Forum was held in Paris on November 10, 1995. The scheduled presentations included "Recent Developments in the Satellite Operator and Services Market" by Mr. K. Madders, "Recent Developments in the Space Insurance Market" by Mr. M. Ramadoro, and additional issues suggested by the participants.

"Doing Business in Outer Space: Global Navigation Satellite System (GNSS)" was the topic of discussion at a conference organized by the U.S. Department of Transportation in cooperation with the Georgetown University Law Center Space Law Group on November 16-17, 1995.

Brief News

Recent **Hubble** observations suggest that the universe may be younger than some of its stars. Astronomers, using a space telescope, have reportedly detected the primordial gas that suffuses all of space and from which all things in the universe were formed....Swiss astronomers claim to have discovered the first planet outside Earth's solar system, revolving around a star in the constellation Pegasus.

Free access to the vast holdings in the **NASA** STI database is available via the Internet....The General Accounting Office estimates that NASA's space station will cost $94 billion....NASA cut off communications with its **Pioneer 11** spacecraft, which has been exploring the solar system for 22 years, on September 30, 1995 due to insufficient power supply....NASA has commissioned a study to explore what private and governmental actions are necessary in order to expedite a large space tourism enterprise....NASA plans to launch a robotic spacecraft sometime in 1999 that will collect dust from a comet and return it to Earth for study.

Thanks to **Cospas/Sarsat**, a satellite-based distress location system, 4,535 people have been rescued in the course of 1,502 events since

1982....In August 1995 in Windsor, Ontario, a vacant mobile home was reportedly destroyed when a chunk of a meteor slammed into it....Two U.S. projects are under development to detect near-earth objects (comets or asteroids) that could collide with the earth.

The international space station's U.S. lab will be covered by a material similar to that of a bulletproof vest to protect it from space debris.

Ulysses, a joint project of NASA and ESA, completed its first orbit around the sun on September 29....The **Galileo** spacecraft is to arrive at its Jupiter orbit on December 7, 1995.

Five NASA astronauts and two Russian cosmonauts were carried by the space shuttle **Atlantis** to the Russian space station **Mir** in July 1995, one of the first of many dockings which are planned before November 1997, when the U.S., Russia and other countries begin construction of the international space station. The docking with Mir was flawless. It was the first link-up of U.S. and Russian spacecraft since the Apollo-Soyuz docking two decades ago.

A Russian **Soyuz** rocket carried a German astronaut and two Russian cosmonauts to the Mir space station in September 1995....A NASA astronaut planning to spend time aboard the Mir space station has been pulled from training because the 6-foot-2 astronaut is just over the Russian space agency's height requirement for the Soyuz capsules.

On Nov. 15, **Atlantis** docked again with Mir, this time with a 15-foot docking tunnel, which was left behind on Mir to make the subsequent dockings easier to accomplish. The combined crew in the Mir consisted of 5 astronauts and 3 cosmonauts representing four countries - the U.S., Russia, Germany, and Canada; this constitutes a record for a single spacecraft.

A two-ton orbiting **Chinese** satellite was reported out of control on November 28 and is expected to crash possibly on the continental U.S. around April 1, 1996.

Meeting in France from October 18-20, the member governments of the **ESA** agreed to finance Europe's participation in the international space station. The price-tag for the years 1996-2004 is estimated to be the equivalent of $3.9 billion.

Globalstar, a low-earth orbiting satellite-based digital telecommunications system that will offer wireless telephone and other telecommunications services world-wide, expects to begin launching satellites in the second half of 1997 and offer initial commercial operations via a 24-satellite constellation in 1998. Reportedly, the **1995 World Radio Conference** has allocated the necessary feeder link frequencies which, together with the user link frequencies allocated in 1992, should complete the spectrum allocation requirements for the Globalstar system....The establishment of **WorldTel**, a private multi-national investment and development organization, was announced at ITU Headquarters in Geneva.

The National Space Development Agency of **Japan** (NASDA) launched TR-IA rocket No. 4 on August 25, 1995. The major purpose of the